

# A Counterexample to Conjecture 00000000269: the Galois Group of $x^n + x + 1$ Is Not $S_n$ When $n \equiv 2 \pmod{3}$

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## Abstract

Conjecture 00000000269 asserts that for all sufficiently large  $n$  the Galois group of  $x^n + x + 1$  is the symmetric group  $S_n$ . It is false: for every  $n \equiv 2 \pmod{3}$  the trinomial is divisible by  $x^2 + x + 1$ , and its Galois group then has at most  $2(n-2)! < n!$  elements, so it is not even abstractly isomorphic to  $S_n$ . Such  $n$  are unbounded, so no threshold works. The refutation is formalized in Lean 4 against Mathlib.

## 1 The conjecture as stated

The original text of conjecture 00000000269 reads:

Definition: The Galois group of  $x^n + x + 1$ . Conjecture: For all sufficiently large  $n$ , this group is  $S_n$  (Selmer proved the corresponding statement for  $x^n - x - 1$ ).

## 2 Reading of the statement

The polynomial  $f_n(x) = x^n + x + 1$  has integer coefficients, and its Galois group is taken over  $\mathbb{Q}$ , as in the work of Selmer to which the statement refers.  $\text{Gal}(f_n)$  denotes the automorphism group of a splitting field of  $f_n$  over  $\mathbb{Q}$ .

“This group is  $S_n$ ” is usually understood as:  $\text{Gal}(f_n)$ , acting on the  $n$  roots of  $f_n$ , is the full symmetric group on them. Any such reading implies the weakest one,

$$\text{Gal}(f_n) \cong S_n \quad \text{as abstract groups,} \tag{1}$$

and this is the statement we refute. So the conjecture asserts that there is an  $N$  such that (1) holds for every  $n \geq N$ , and we show that (1) fails for infinitely many  $n$ .

## 3 Result

**Theorem 1.** *Let  $n \geq 5$  with  $n \equiv 2 \pmod{3}$ . Then  $|\text{Gal}(f_n)| \leq 2(n-2)! < n!$ . In particular  $\text{Gal}(f_n) \not\cong S_n$ , and Conjecture 00000000269 is false.*

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## 4 Proof

**Lemma 2.** *If  $n = 3k + 5$  with  $k \geq 0$ , then  $x^2 + x + 1$  divides  $x^n + x + 1$  in  $\mathbb{Q}[x]$ .*

*Proof.* We have the identity

$$x^{3k+5} + x + 1 = x^2((x^3)^{k+1} - 1) + (x^2 + x + 1).$$

Since  $x^3 - 1$  divides  $(x^3)^{k+1} - 1$  and  $x^3 - 1 = (x - 1)(x^2 + x + 1)$ , both terms on the right are divisible by  $x^2 + x + 1$ . (Equivalently: a primitive cube root of unity  $\omega$  satisfies  $\omega^n + \omega + 1 = \omega^2 + \omega + 1 = 0$ .)  $\square$

**Lemma 3.** *If  $p \in \mathbb{Q}[x]$  has degree  $d$ , then  $|\text{Gal}(p)| \leq d!$ .*

*Proof.*  $\text{Gal}(p)$  permutes the roots of  $p$  in its splitting field  $L$ , and this action is faithful because  $L$  is generated over  $\mathbb{Q}$  by those roots. There are at most  $d$  roots, so  $\text{Gal}(p)$  embeds in a symmetric group on at most  $d$  letters.  $\square$

**Lemma 4.** *For  $p, q \in \mathbb{Q}[x]$ ,  $|\text{Gal}(pq)| \leq |\text{Gal}(p)| \cdot |\text{Gal}(q)|$ .*

*Proof.* The splitting field  $M$  of  $pq$  contains splitting fields of  $p$  and of  $q$ , and every automorphism of  $M$  maps each of them to itself. Restriction therefore gives a homomorphism  $\text{Gal}(pq) \rightarrow \text{Gal}(p) \times \text{Gal}(q)$ . It is injective, because  $M$  is generated by the roots of  $p$  and of  $q$ , so an automorphism of  $M$  that fixes both subfields is the identity.  $\square$

*Proof of Theorem 1.* Write  $n = 3k + 5$ . By Lemma 2,  $f_n = (x^2 + x + 1)g$  for some  $g \in \mathbb{Q}[x]$ , and comparing degrees gives  $\deg g = n - 2$ . By Lemmas 4 and 3,

$$|\text{Gal}(f_n)| \leq |\text{Gal}(x^2 + x + 1)| \cdot |\text{Gal}(g)| \leq 2!(n - 2)! = 2(n - 2)!.$$

On the other hand  $n! = n(n - 1)(n - 2)!$  and  $n(n - 1) \geq 20 > 2$ , so  $|\text{Gal}(f_n)| < n! = |S_n|$  and (1) fails.

Given any threshold  $N$ , the integer  $n = 3N + 5$  satisfies  $n \geq N$ ,  $n \geq 5$  and  $n \equiv 2 \pmod{3}$ , so (1) fails for it. Hence no threshold has the property asserted by the conjecture.  $\square$

## 5 Formalization

The accompanying Lean 4 project (`lean/Submission/Basic.lean`) states the conjecture and proves its negation against Mathlib, whose `Polynomial.Gal p` is the Galois group of the splitting field of  $p$ .

- `trinomials n` — the polynomial  $x^n + x + 1 \in \mathbb{Q}[x]$ .
- `GalIsSymmetric n` — (1): a group isomorphism  $(\text{trinomials } n).\text{Gal} \simeq^* \text{Equiv.Perm}(\text{Fin } n)$  exists.
- `ConjectureHolds` — there is  $N$  with `GalIsSymmetric n` for all  $n \geq N$ .
- `card_gal_le` — Lemma 3, from Mathlib's `Gal.galActionHom_injective` and `ncard_rootSet_le`.
- `card_gal_mul_le` — Lemma 4, from Mathlib's `Gal.restrictProd_injective`.
- `cyclotomic_dvd_trinomial` — Lemma 2; `natDegree_cyclotomic`, `natDegree_trinomial` — the degrees 2 and  $n$ .

- `card_gal_trinomial_le` —  $|\text{Gal}(f_{3k+5})| \leq 2(3k+3)!$ ; `not_galIsSymmetric` — (1) fails for  $n = 3k + 5$ .
- `conjecture_00000000269_false` —  $\neg \text{ConjectureHolds}$ , by taking  $n = 3N + 5$ .

The project builds with `lake build` on `leanprover/lean4:v4.33.1` with `Mathlib v4.33.1`. It contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms: `#print axioms conjecture_00000000269_false` reports exactly `propext`, `Classical.choice` and `Quot.sound`.

## 6 Remarks

The divisibility in Lemma 2 is classical. Selmer [1] showed that  $x^n + x + 1$  is irreducible over  $\mathbb{Q}$  for  $n \not\equiv 2 \pmod{3}$ , and that for  $n \equiv 2 \pmod{3}$ ,  $n > 2$ , it factors as  $(x^2 + x + 1)g_n(x)$  with  $g_n$  irreducible. So the obstruction found here is the only reducibility of the family, and it occurs for one residue class in three. The conjecture could be repaired by restricting to  $n \not\equiv 2 \pmod{3}$ , or by replacing  $x^n + x + 1$  with  $g_n$  when  $n \equiv 2 \pmod{3}$ ; we do not address either version. The statement Selmer proved for  $x^n - x - 1$ , which the conjecture cites, is irreducibility for every  $n$ . That polynomial has no such cyclotomic factor.

## References

- [1] E. S. Selmer, *On the irreducibility of certain trinomials*, Math. Scand. 4 (1956), 287–302.